

02 — BiCRS and Bio-fuel: The Chemistry

The chemical pathways of biomass carbon removal and storage, inextricably linked to the production of bio-ethanol and bio-methanol.

What this instalment explores

- BiCRS — biomass carbon removal and storage — as a significant category within the negative emission technologies recognised by the IPCC.
- Three distinct pathways for Juncao biomass: direct gasification with CO₂ injection into geological storage, biomass-to-liquids with integrated CO₂ capture, and the fermentation into ethanol alongside the capture of fermentation-derived CO₂.
- Bio-methanol via the syngas pathway: providing a higher energy density and proving itself suitable for the exacting requirements of shipping and aviation.
- Cost curves: the current LCOE of BiCRS pathways in comparison with natural gas, kerosene, and methanol derived from natural gas.
- The role of European chemical faculties and process technology — including TU Delft, Aachen, Stuttgart, and Lyon — which produce leading publications that are, regrettably, seldom translated into physical industrial installations.
- The Asian lead: Chinese consortia that are applying these pathways commercially on a smaller scale and effectively exploiting the knowledge yield of Lin Zhanxi's circle.

This instalment is currently being further refined with process diagrams, energy balances per tonne of biomass, and an analysis of the three distinct end products that find an immediate market in Europe.